MidTransport

Open-Source, Privacy-First NEMT SaaS Platform

Research & Build Proposal

> **Status:** Under Review **Prepared:** March 2026 **Stack Confirmed:** Node.js / TypeScript · React · PostgreSQL · Docker **MVP Target:** Small NEMT provider (1–20 vehicles)

Executive Summary

MidTransport is an open-source, self-hostable Non-Emergency Medical Transportation (NEMT) platform for small providers serving sick and elderly patients. Unlike every competing product on the market today, MidTransport stores all patient and trip data locally — on the provider's own infrastructure — eliminating data sharing with competitors and addressing the #1 privacy concern raised by NEMT providers.

Core differentiators:

1. **Local-first / Self-hostable** — your data stays on your server
2. **Open-source (MIT)** — auditable, no vendor lock-in
3. **Privacy by design** — HIPAA compliant, field-level PHI encryption
4. **Multiple data entry methods** — manual, CSV upload, broker APIs
5. **Accessible UI** — mobile-first for drivers, large-text for elderly patients
6. **No per-trip fees** — self-hosted is free, managed cloud is flat-rate

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1. Market Opportunity

Metric	Data
US NEMT Market Size (2025)	~$4.4–6.6 billion
Projected US Market (2031)	~$13.4 billion
CAGR	7.1% – 9.7%
Medicaid share of NEMT trips	52.34%
Largest player market share	< 5% (highly fragmented)
Fastest growing segment	Mental health transport (10.11% CAGR)
Healthcare data breaches (2023)	809 breaches / 133M records exposed
HIPAA max fine	$1.5M/year per violation category

The Gap: No dominant open-source, self-hostable NEMT platform exists. Every major competitor is a closed SaaS cloud product. Privacy concerns are rising. This is the market white space.

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2. Competitive Analysis

Top NEMT Software Platforms in the USA

Product	Pricing	Key Strength	Key Weakness	Self-Hostable?
MediRoutes	Not public	Open API since 2008, broker integrations	Closed SaaS	No
RouteGenie	$50/vehicle/mo	Route optimization (-10–20% miles)	Complicated UI	No
TripMaster	From $125/mo	Scheduling automation, GPS	Basic route optimization	No
Tobi Cloud	Quote-based	Easy UI, digital attestation	Limited reporting	No
NEMT Platform	Not public	AI dispatch, broker API integrations	Closed SaaS	No
Momentm Technologies	Enterprise	30yr expertise (Trapeze/TripSpark)	Enterprise pricing only	No
Upper	SaaS tiers	Good for small providers	NEMT features limited	No
Ecolane	Enterprise	Public transit + NEMT	Not NEMT-only	No
Bambi NEMT	SaaS	HIPAA-focused	Smaller ecosystem	No
RydeWizz	Free	Free tier available	Cloud-only, limited features	No
MidTransport	Free (self-host)	Privacy-first, open source	New — needs community	**Yes**

Privacy Concerns Documented

- All major platforms run shared cloud infrastructure — providers fear competitors can access their patient lists
- Business Associate Agreements (BAAs) required under HIPAA but inconsistently enforced
- 2023: Record 809 healthcare data breaches exposed 133M records
- New HCPCS S0215 compliance requirement effective July 1, 2025
- Providers in smaller markets report reluctance to adopt cloud software for this reason

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3. Available Broker APIs & Integrations

Major NEMT Brokers (Trip Sources)

Broker	Coverage	API Integration

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ModivCare (formerly LogistiCare)	30+ states	REST API · Used by RouteGenie, NEMT Platform
MTM Inc.	All 50 states (post Access2Care acquisition)	MTM Link API · Real-time trip import
Kaiser Permanente	West Coast	API via NEMT Platform
Call the Car	California	API available
MAS (Medical Answering Services)	Regional	REST API · Used by MediRoutes

MediRoutes API (Reference / Benchmark)

MediRoutes has published an open API since 2008:

- Real-time trip data: scheduled times, en-route, arrival, performance, no-shows, signatures, costs
- Automated scheduling: 75% of trips scheduled automatically via API rules
- Broker portals push trips directly — eliminates manual data entry
- Saves ~2–3 hours/day of administrative work per dispatcher

MidTransport Integration Priority (Phase 2):

1. CSV import — works with every broker immediately (all export CSV)
2. MTM Link API — largest broker network (all 50 states)
3. ModivCare API — most trips nationally
4. MAS API — medical answering service integration
5. Manual entry — always available, no API required

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4. Unique Value Propositions

Value Prop	MidTransport	All Competitors
Self-hostable	Yes	No
Open source	Yes (MIT)	No
Patient data stays on your server	Yes	No
CSV import	Yes	Most
Manual data entry	Yes	Most
Broker API integration	Phase 2	Yes
Free to run	Yes (self-host)	No (all paid)
Mobile driver app	Yes	Yes
Accessible UI (elderly)	Yes	Partial

Value Prop	MidTransport	All Competitors
HIPAA by design	Yes	SaaS BAA only

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5. Feature Roadmap

Phase 1 — MVP (Weeks 1–8)

Core operational capability for a small NEMT provider running their own server.

Patient Management

- ☐ Create / edit / archive patients
- ☐ Medical notes, mobility requirements (wheelchair, stretcher, etc.)
- ☐ CSV bulk import (drag-and-drop + column mapping)
- ☐ Duplicate detection

Trip Scheduling

- ☐ Manual trip entry form
- ☐ Recurring trip templates
- ☐ CSV trip import
- ☐ Trip status workflow: Scheduled → Dispatched → En Route → Arrived → Completed → Billed

Driver Management

- ☐ Driver profiles, license, vehicle assignment
- ☐ Availability / shift scheduling
- ☐ Trip assignment (manual or auto-assign)

Dispatcher Dashboard (Web)

- ☐ Live trip board (kanban-style by status)
- ☐ Map view (Leaflet + OpenStreetMap — free)
- ☐ Search / filter trips
- ☐ Notifications panel (alerts, no-shows)

Driver Mobile App (PWA)

- ☐ Today's trips list
- ☐ One-tap navigation (opens Google Maps / Apple Maps)
- ☐ Status update buttons (En Route / Arrived / Completed)
- ☐ Digital signature capture

Notifications

- ☐ SMS to driver on new assignment (Twilio)
- ☐ SMS/email to patient with trip reminder
- ☐ Email to dispatcher on trip completion

Reporting

- ☐ Trips per day / week / month
- ☐ Driver utilization
- ☐ On-time performance
- ☐ Export to CSV

Security & Compliance

- ☐ Role-based access: Admin · Dispatcher · Driver · Facility Coordinator
- ☐ HIPAA audit log (all PHI access logged with timestamp + user)
- ☐ Field-level encryption for PHI (name, DOB, address, medical notes)
- ☐ TLS enforced on all connections
- ☐ Password hashing (bcrypt)
- ☐ JWT authentication with refresh tokens

Phase 2 — Integrations (Weeks 9–16)

- ☐ Route optimization (OSRM — self-hosted, free)
- ☐ ModivCare API integration (real-time trip import)
- ☐ MTM Link API integration
- ☐ Real-time GPS tracking (driver mobile app)
- ☐ Billing module (Medicaid claim format: 837P)
- ☐ Patient self-service portal (book / cancel trips)
- ☐ Digital attestation (driver sign-off)
- ☐ Advanced analytics dashboard
- ☐ Multi-provider support (one server, multiple orgs)

Phase 3 — Scale (Weeks 17–24)

- ☐ AI-assisted auto-scheduling
- ☐ Insurance eligibility verification API
- ☐ EHR/EMR integration (HL7 FHIR R4)
- ☐ Voice interface for elderly patients (Web Speech API)
- ☐ Fleet maintenance tracking
- ☐ React Native mobile apps (App Store + Google Play)
- ☐ Managed cloud tier (for providers who prefer hosted)
- ☐ Community marketplace for plugins

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6. Recommended Tech Stack

Full-Stack TypeScript Monorepo

```
midtransport/
├── apps/
│   ├── web/          # React 18 + TypeScript (dispatcher dashboard)
│   ├── mobile/       # React Native + Expo (driver app)
│   └── api/          # Node.js + Express + TypeScript (REST API)
```

```
  ├── packages/
  │   └── shared/       # Shared types, utils, validation schemas
  └── docker-compose.yml
```

Technology Choices

Layer	Choice	Reason
Frontend framework	React 18 + TypeScript	Mature, large ecosystem, best for dashboards
UI components	shadcn/ui + TailwindCSS	Accessible (WCAG 2.1 AA), highly customizable
Data fetching	TanStack Query	Caching, background updates, optimistic UI
Maps	Leaflet + OpenStreetMap	Free, self-hostable, no API key needed
Mobile	React Native (Expo)	Code sharing with web, iOS + Android
Backend	Node.js + Express + TypeScript	Single language across full stack
Database	PostgreSQL	HIPAA-ready, mature, self-hostable
Cache / Realtime	Redis + Socket.io	Live trip status updates
Routing engine	OSRM (Phase 2)	Open source, self-hostable, fast
Auth	JWT + bcrypt	Standard, secure, stateless
Containerization	Docker + Docker Compose	One-command deploy on any Linux server
SMS/Email	Twilio + SendGrid	Reliable, affordable at small scale

Self-Hosting Target

Runs on a **$10–15/month VPS** (DigitalOcean, Hetzner, Vultr):

- 2 vCPU, 4GB RAM — handles 5–10 concurrent dispatchers, 50+ drivers
- Single `docker-compose up -d` to deploy everything

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7. UI/UX Design Approach

User Roles & Primary Screens

Role	Device	Key Screens
Admin	Desktop	Dashboard · Users · Settings · Reports
Dispatcher	Desktop	Live Trip Board · Map · Notifications · Schedule
Driver	Mobile	Trip List · Active Trip · Navigation · Signature
Patient/Rider	Mobile / tablet	Book Trip · My Trips · Cancel
Facility Coordinator	Desktop / tablet	CSV Upload · Schedule Patients · Status

Accessibility Standards (Elderly Patients)

- Minimum 16px font (18–20px preferred)
- High contrast mode — WCAG AA (4.5:1 contrast ratio minimum)
- Touch targets minimum 48×48px
- 2–3 step booking flow maximum
- Voice input support (Web Speech API)
- Full screen reader support (ARIA labels, roles, live regions)
- No small icons without text labels

Dispatcher Dashboard Wireframe

```
┌──────────────────────────────────────────────────────────────────┐
│  MidTransport  |  Mon Mar 10, 2026  |  🔔 3 Alerts  |  👤        |
├─────────────────┼──────────────────────────────┼──────────────────┤
│  NAVIGATION     |  LIVE TRIP BOARD             |  MAP             |
│                 |  [Search...] [+ Add Trip]    |  (OpenStreetMap) |
│  📊 Dashboard   |  [Filter: All ▼]             |                  |
│  🚐 Trips       |                              |  🚐 Driver pins  |
│  👥 Patients    |  ┌──────────────────────┐    |  📍 Pickup pins  |
│  🧑 Drivers     |  | ⚪ SCHEDULED         |    |  🏥 Drop pins    |
│  📅 Schedule    |  | Maria Rodriguez      |    |                  |
│  📈 Reports     |  | Pickup: 9:30am       |    |  [Zoom In/Out]   |
│  ⚙️ Settings    |  | Driver: James K.     |    |  [Full Screen]   |
│                 |  | → St. Mary's Hosp.   |    |                  |
│                 |  | [Assign] [Cancel]    |    |                  |
│                 |  └──────────────────────┘    |                  |
│                 |  [CSV Import]                |                  |
└─────────────────┴──────────────────────────────┴──────────────────┘
```

Driver Mobile App Wireframe

```
┌──────────────────────┐
│  Today's Trips  (5)  |
│  Mon Mar 10, 2026    |
├──────────────────────┤
│  ▶  NEXT UP          |
│  ────────────────    |
│  Maria Rodriguez     |
│  Pickup: 9:30 AM     |
│  📍 412 Oak St.      |
│  → St. Mary's Hosp.  |
│                      |
│  ┌──────┐ ┌────────┐ |
│  | NAV  | |ARRIVED | |
│  └──────┘ └────────┘ |
├──────────────────────┤
│  10 Trip 3  11:00am  |
│  11 Trip 4   1:30pm  |
```

```
│  12 Trip 5   3:00pm  │
└──────────────────────┘
```

CSV Import Flow

1. **Upload** — Drag-and-drop zone accepts `.csv` files up to 50MB (~10,000 rows)
2. **Preview** — First 5 rows shown; errors highlighted in red
3. **Map Columns** — UI to map CSV headers → system fields (patient name, DOB, address, etc.)
4. **Validate** — Duplicate detection, required field check, date format validation
5. **Import** — Progress bar; summary report (imported / skipped / errors)
6. **Template** — Download pre-formatted CSV template to fill in

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8. Project File Structure

```
midtransport/
├── docker-compose.yml              # Postgres + Redis + API + Web
├── docker-compose.dev.yml          # Dev overrides (hot reload)
├── .env.example                    # All environment variables documented
├── apps/
│   ├── web/                        # React dispatcher dashboard
│   │   ├── src/
│   │   │   ├── pages/
│   │   │   │   ├── Dashboard.tsx
│   │   │   │   ├── Trips.tsx
│   │   │   │   ├── Patients.tsx
│   │   │   │   ├── Drivers.tsx
│   │   │   │   └── Reports.tsx
│   │   │   └── components/
│   │   │       ├── TripBoard/
│   │   │       ├── MapView/        # Leaflet + OpenStreetMap
│   │   │       └── CSVImport/      # Drag-drop + column mapping
│   │   └── package.json
│   ├── mobile/                     # React Native driver app
│   │   └── src/screens/
│   │       ├── TripList.tsx
│   │       └── ActiveTrip.tsx
│   └── api/                        # Node.js + Express backend
│       ├── src/
│       │   ├── routes/
│       │   │   ├── auth.ts
│       │   │   ├── trips.ts
│       │   │   ├── patients.ts
│       │   │   ├── drivers.ts
│       │   │   └── import.ts       # CSV import endpoint
│       │   ├── db/
│       │   │   └── schema.sql      # PostgreSQL schema
│       │   ├── middleware/
│       │   │   ├── auth.ts         # JWT validation
│       │   │   └── audit.ts        # HIPAA audit log
```

```
│       │   └── integrations/       # Phase 2
│       │       ├── mtm.ts
│       │       └── modivcare.ts
│       └── package.json
├── packages/
│   └── shared/
│       └── src/types/              # Shared TypeScript types
└── docs/
    ├── BAA-template.md             # Business Associate Agreement template
    ├── HIPAA-compliance.md         # How MidTransport achieves HIPAA compliance
    ├── self-hosting.md             # Deployment guide
    └── csv-import-template.csv     # Template for CSV data entry
```

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9. Implementation Estimate

Phase	Scope	Sessions	Est. Tokens	Est. API Cost
Phase 1 — Foundation	Docker, DB, Auth, CRUD, Dashboard, PWA, Maps, CSV	9–11 sessions	~930K	~$8–12
Phase 2 — Integrations	Broker APIs, Routing, GPS, Billing, Patient Portal	7–10 sessions	~850K	~$7–10
Phase 3 — Scale	Multi-tenant, AI, FHIR, Mobile Apps, Cloud Tier	9–10 sessions	~950K	~$8–12
Total		~25–30 sessions	~2.7M	~$23–35

Each "session" = one Claude Code conversation. Sessions start fresh with context handed over. **Phase 1 alone = fully working, shippable MVP for a small NEMT provider.**

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10. Sources

- NEMT Market Report 2025 — NEMTrepreneur
- NEMT Market $16.24B by 2032 — BusinessWire
- Top 10 NEMT Software 2025 — NEMTrepreneur
- NEMT Software Comparison — UpperInc
- MediRoutes API Documentation
- Broker Integrations Guide — NEMT Platform
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- Healthcare App UX Best Practices — Eleken
- Accessibility for Elderly — PMC
- NEMT Industry Statistics 2026 — Elite Med Financials
- Healthcare Data Security 2025 — Healthcare IT Today